

The Net Change Theorem.

Part 2 of the fundamental thm says that if F is a continuous function on $[a, b]$ then

$$\int_a^b f(x) dx = F(b) - F(a).$$

We know (from 2.2.4) that $F'(x)$ represents the rate of change of $y = F(x)$ with respect to x and $F(b) - F(a)$ is the change in y when x changes from a to b .

Net change thm. The integral of a rate of change is the net change

$$\int_a^b F'(t) dt = F(b) - F(a).$$

- If $V(t)$ is the volume of water in reservoir at time t , then its derivative $V'(t)$ is the rate at which water flows into the reservoir at the time t , so.

$$\int_{t_1}^{t_2} V'(t) dt = V(t_2) - V(t_1).$$

is the change in the amount of water in the reservoir between time t_1 and t_2 .

- If the mass of a rod measured from the left end to a point x is $m(x)$, then the linear density is $\rho(x) = m'(x)$ so,

$$\int_a^b \rho(x) dx = m(b) - m(a).$$

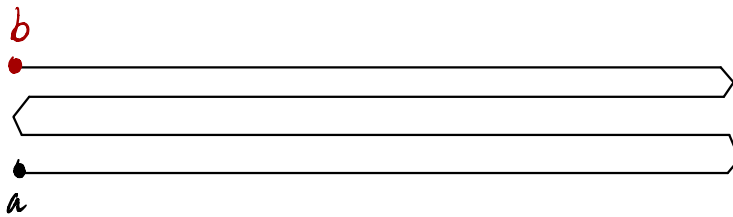
is the mass of the segment of the rod that lies between $x=a$ and $x=b$.

- If an object moves along a straight line with position function $s(t)$, then its velocity is $v(t) = s'(t)$ so,

$$\int_{t_1}^{t_2} v(t) dt = s(t_2) - s(t_1).$$

is the net change of position or displacement of the object during the time period from t_1 to t_2 .

Rmk: If we want to calculate the distance the object travel during a time interval we have to consider the intervals when $v(t) \geq 0$ (the object moves to the right) and also the intervals when $v(t) \leq 0$ (the object moves to the left).



In both cases the distance is computed by integrating $|v(t)|$ the speed. Therefore,

$$\int_{t_1}^{t_2} |v(t)| dt = \text{total distance traveled.}$$

Example. A particle moves along a line so that its velocity at time t is $v(t) = t^2 - t - 6$

a). Find the displacement of the particle during the time period $1 \leq t \leq 4$.

b). Find the distance traveled during this time period.

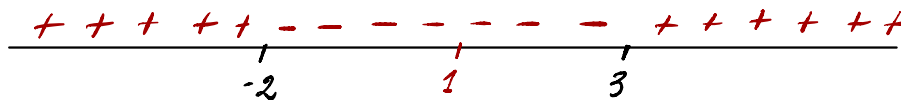
Sln. a). By $v(t) = t^2 - t - 6$.

$$s(4) - s(1) = \int_1^4 v(t) dt = \int_1^4 (t^2 - t - 6) dt = \left. \frac{t^3}{3} - \frac{t^2}{2} - 6t \right|_1^4$$

$$S(4) - S(1) = \frac{4^3}{3} - \frac{4^2}{2} - 6(4) - \frac{1}{3} + \frac{1}{2} + 6 = -\frac{9}{2}.$$

this means that particle moved 4.5 m toward the left.

b). Notice that $v(t) = t^2 - t - 6 = (t-3)(t+2)$ So.



then $v(t) \geq 0$ on $[3, 4]$ and $v(t) \leq 0$ $[1, 3]$.

So,

$$\begin{aligned} d &= \int_1^4 |v(t)| dt = \int_1^3 |v(t)| dt + \int_3^4 |v(t)| dt \\ &= \int_1^3 (-v(t)) dt + \int_3^4 v(t) dt \\ &= \int_1^3 (-(t^2 - t - 6)) dt + \int_3^4 (t^2 - t - 6) dt \\ &= \int_1^3 (-t^2 + t + 6) dt + \int_3^4 (t^2 - t - 6) dt \\ &= \left\{ -\frac{t^3}{3} + \frac{t^2}{2} + 6t \right\} \Big|_1^3 + \left\{ \frac{t^3}{3} - \frac{t^2}{2} - 6t \right\} \Big|_3^4 \\ &= \frac{61}{6} \approx 10.17 \text{ m.} \end{aligned}$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$